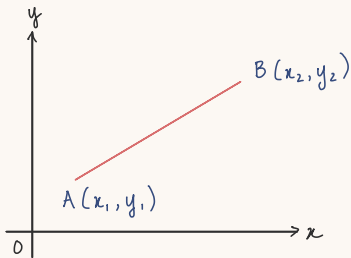


PURE 3. Coordinate Geometry

1) Straight lines

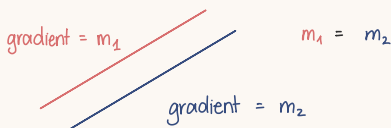


• gradient of AB = $\frac{y_2 - y_1}{x_2 - x_1}$ ← $\frac{\text{rise}}{\text{run}}$

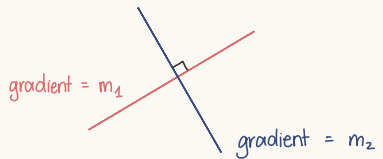
• length of AB = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Pythagoras' theorem

• midpoint of AB = $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

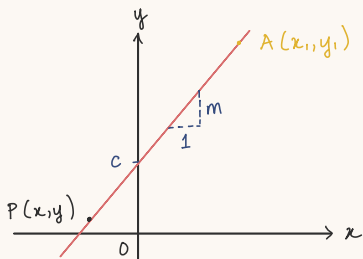
* parallel lines → same gradient



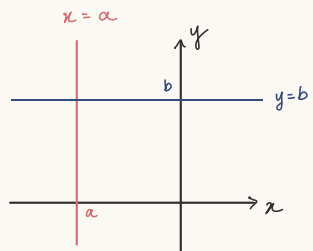
* perpendicular lines :



$m_1 \times m_2 = -1$; $m_2 = -\frac{1}{m_1}$



$y = mx + c$
y - intercept
gradient



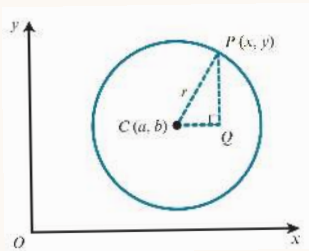
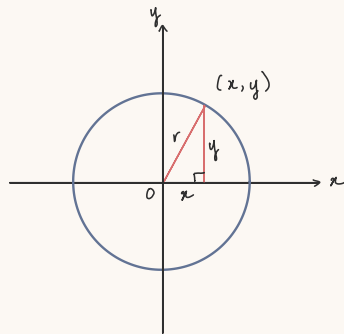
• gradient m , pass through point (x_1, y_1)

↳ equation: $y - y_1 = m(x - x_1)$

2. Circles

- center (x, y)
- radius r

$$x^2 + y^2 = r^2$$



- center (a, b)

- radius r

↳ equation of a circle:

1) completed square form

$$(x-a)^2 + (y-b)^2 = r^2$$

Expanding $(x-a)^2 + (y-b)^2 = r^2$ gives:

$$x^2 - 2ax + a^2 + y^2 - 2by + b^2 = r^2$$

Rearrange $\rightarrow x^2 + y^2 - 2ax - 2by + (a^2 + b^2 - r^2) = 0$

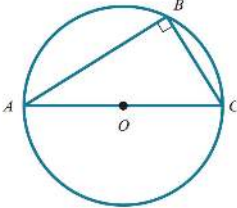
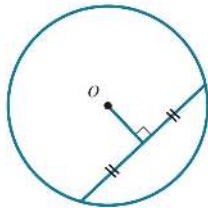
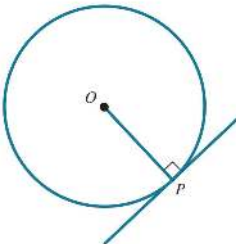
↳ 2) equation of circle in expanded form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

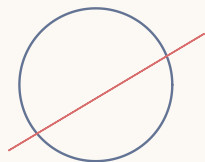
- center $(-g, -f)$

- radius $\sqrt{g^2 + f^2 - c}$

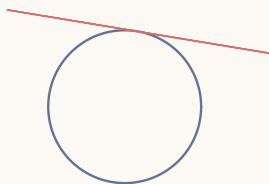
3. some right angle facts

		
The angle in a semicircle is a right angle.	The perpendicular from the centre of a circle to a chord bisects the chord.	The tangent to a circle at a point is perpendicular to the radius at that point.

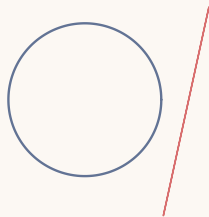
4. Intersections of lines and circles



$$b^2 - 4ac > 0$$



$$b^2 - 4ac = 0$$



$$b^2 - 4ac < 0$$

$b^2 - 4ac$	Nature of roots	Line and parabola
> 0	two distinct real roots	two distinct points of intersection
$= 0$	two equal real roots	one point of intersection (line is a tangent)
< 0	no real roots	no points of intersection